

Quarterly equity audit protocol for blockchain-logged traffic signal AI

§1 Scope and data model

This chapter specifies a reproducible, quarterly equity audit protocol that can be executed entirely against simulation output from Eclipse SUMO, with no access to real-world traffic data or personally identifiable information. The protocol answers a single question: **does the AI signal controller distribute delay equitably across neighbourhoods of different deprivation levels and ethnic compositions, and does it do so consistently over successive model versions?**

What the audit covers and what it does not

The audit covers every phase-selection decision logged by the AI controller during a SUMO simulation episode. It evaluates distributional fairness of per-vehicle delay aggregated to Lower Layer Super Output Areas (LSOAs), joined to the English Indices of Multiple Deprivation 2019 (IMD 2019) and ONS 2021 Census ethnic group data. It does not cover real traffic, real travellers, or real demographic exposure. All findings are conditional on SUMO's traffic model fidelity and the demand scenarios fed to it. The audit cannot detect induced non-travel, mode shift, or any harm that does not manifest as measurable vehicle or pedestrian delay inside the simulator.

The per-decision log schema

Every AI phase-selection event writes a record to an append-only blockchain-anchored log with the following fields: **timestamp** (simulation-clock, millisecond resolution), **intersection ID** (SUMO junction identifier, one-to-one with the OSM node it was derived from), **sensor state hash** (SHA-256 of the detector occupancy vector, enabling tamper detection), **phase ID action** (single-token identifier of the selected signal phase), **audit-path chain-of-thought trace** (the model's internal reasoning serialised as a JSON object, enabling post-hoc interpretability review), **urgency scores** (per-approach floating-point values the model assigned to competing traffic streams), and **model version** (semantic version string, incremented at every retraining event). The intersection ID field enables spatial joins to LSOAs; the timestamp enables temporal stratification; urgency scores enable detection of systematic priority bias; the model version field partitions the log into auditable epochs aligned with the quarterly cadence.

Why the Elephant & Castle-Brixton corridor

The worked example throughout §3 uses a **5 × 5 km section of inner south London straddling the Lambeth-Southwark borough boundary**, running approximately from Elephant & Castle in the north to Brixton in the south. This area is selected for three reasons. First, **extreme IMD variance in close proximity**: Southwark's Joint Strategic Needs Assessment confirms LSOAs ranging from IMD decile 1 (most deprived, notably East Walworth and South Bermondsey) through to deciles 8-9 (Dulwich), while Lambeth spans a comparable range from Brixton and

Stockwell estates to Herne Hill and Clapham (MHCLG, 2019; Southwark JSNA Factsheet). Second, the corridor is **TfL-managed with dense signalised junctions** on the A23, A3, and A202, providing a realistic test of AI signal control at scale. Third, the ONS 2021 Census shows **high ethnic diversity** at LSOA level, enabling meaningful disaggregation by the five high-level ethnic group categories (ONS, 2022).

§2 Component surveys

§2.1 Fairness metrics computable from per-decision logs

Ten candidate metrics were evaluated for computability from SUMO output, sensitivity to the distributional patterns of interest, and precedent in the traffic-signal or algorithmic-fairness literature. For each, the unit of observation is mean per-vehicle delay at the LSOA level, denoted $\bar{d}_i$ for LSOA i .

Gini coefficient of delay across LSOAs. Defined as $G = \frac{1}{2n^2\bar{d}} \sum_i \sum_j |\bar{d}_i - \bar{d}_j|$, where n is the number of LSOAs and $\bar{d}$ the grand mean delay (Dorfman, 1979). Strengths: scale-invariant, well-understood, decomposable via Lorenz curve. Weaknesses: insensitive to which end of the distribution drives inequality; two distributions can share the same Gini. (Wikipedia) Computable from SUMO TripInfo output aggregated to LSOAs. Used as the **headline metric** in this protocol. Transport precedent exists in public-transit accessibility measurement, where Gini values above 0.3 are flagged as concerning (Taylor & Francis Online) and above 0.4 as significant inequality (Rao et al., 2022; Lee, 2016).

Rawlsian max-min (worst-served LSOA's delay). $M = \max_i \bar{d}_i$, or more usefully, the mean delay in the worst-served IMD decile. No formula beyond simple aggregation; grounded in Rawlsian difference principle. Strength: directly captures the experience of the worst-off. Weakness: sensitive to outliers and single-LSOA effects. Fully computable.

Jain's fairness index. $J = \frac{(\sum_i \bar{d}_i)^2}{n \sum_i \bar{d}_i^2}$, bounded $[1/n, 1]$ (Jain, Chiu & Hawe, 1984). Designed for network resource allocation. (arXiv) Strength: intuitive percentage interpretation ($J = 0.9$ means the system is fair to 90 % of users). (Washington University in S...) Weakness: less sensitive than Gini to transfers among middle values. Computable.

****Generalised Gini welfare functions (GGF).**** $W_{\text{GGF}} = \sum_i w_i \bar{d}_{(i)}$, where $\bar{d}_{(i)}$ are sorted ascending and weights w_i are non-increasing, interpolating between utilitarian ($w_i = 1$) and egalitarian ($w_i = \mathbf{1}_{i=1}$) objectives ([ACM JATS, 2025](https://dl.acm.org/doi/10.1145/3749378)). Strength: parameterises the efficiency-equity trade-off. Weakness: requires choosing weight vectors, adding researcher degrees of freedom. Computable; weight sensitivity analysis recommended.

Delay-based fairness. Raeis and Leon-Garcia define this via the CDF of individual vehicle waiting times, penalising configurations where some vehicles experience extreme delay relative to the population ([Raeis & Leon-Garcia, 2021](#)). Strength: individual-level granularity. Weakness: requires per-vehicle data, computationally expensive at scale. Computable from SUMO `traci.vehicle.getAccumulatedWaitingTime()`.

Throughput-based fairness. From the same authors, this extends generalised processor sharing to ensure proportional green time across competing flows. ([arXiv](#)) Strength: directly maps to signal engineering. Weakness: measures flow-level rather than neighbourhood-level equity. Computable from SUMO lane-detector throughput.

Accessibility thresholds. Martens argues the primary transport good is accessibility, not speed, and proposes sufficiency thresholds below which individuals are disadvantaged ([Wikipedia](#)) ([Martens, 2017](#)). Strength: philosophically grounded. Weakness: threshold selection is normative. Partially computable—requires an accessibility model layered on SUMO delay output.

****Disparate impact ratio (80 % rule).**** Adapted from the 1978 Uniform Guidelines on Employee Selection Procedures :antCitation[] {citations="29516f5f-c75b-4053-bf82-fe55ef4bed9f"} ([29 CFR §1607.4D](https://www.ecfr.gov/current/title-29/subtitle-B/chapter-XIV/part-1607)): $DIR = \bar{d}_{\text{least-deprived}} / \bar{d}_{\text{most-deprived}}$, where the ratio is of delay experienced, and $DIR < 0.8$ indicates potential disparate impact. Adapted for algorithmic auditing by Feldman et al. ([2015, KDD](https://dl.acm.org/doi/10.1145/2783258.2783311)). Weerts et al. caution that transplanting the rule across domains involves epistemic trespassing :antCitation[] {citations="76a0e903-b6de-4293-a459-ea8c67073d48"} ([Weerts et al., 2024, FAccT](https://doi.org/10.1145/3630106.3658938)). Computable. Used as a ****triangulation metric**** in this protocol.

Counterfactual fairness. A decision is counterfactually fair if it would be the same had the individual belonged to a different demographic group ([arXiv](#)) ([Kusner et al., 2017, NeurIPS](#)). In observational settings this requires causal modelling with strong assumptions. ([arXiv](#)) However, **SUMO makes counterfactual fairness directly computable**: the researcher can re-run identical demand with altered neighbourhood demographics in the external join, holding all traffic dynamics constant. This is a distinctive methodological advantage of simulation-based auditing.

Pedestrian-specific fairness. NCHRP Research Report 969 documents that US signal timing routinely produces 80-second pedestrian delays against 20-second vehicle delays and proposes pedestrian delay as a co-equal performance measure ([NCHRP, 2022](#)). FELight introduces a fairness penalty that activates when individual waiting time exceeds a threshold ([Du et al., 2024, IEEE T-ITS](#)). Zhang et al. propose a human-centric reward penalising the number of delayed individuals across all modes—pedestrians, vehicle occupants, and transit passengers—tested across Melbourne scenarios ([arXiv](#)) ([Zhang et al., 2026](#)). Pedestrian delay per LSOA is computable

from SUMO's person-simulation module and is used as the **third triangulation metric** in this protocol.

§2.2 Joining SUMO intersection coordinates to deprivation and demographic data

SUMO networks imported from OpenStreetMap via `netconvert` retain georeferencing: the `<location>` element in `.net.xml` stores the UTM projection parameters and offset, and each `<junction>` element carries (x, y) coordinates in that projection ([SUMO Documentation](#)). Converting junction coordinates to WGS84 longitude-latitude is accomplished via `sumolib.net.convertXY2LonLat()` ([SUMO Geo-Coordinates](#)). Each junction's coordinate is then point-in-polygon tested against ONS 2021 LSOA boundary polygons, available as GeoJSON from the ONS Open Geography Portal ([ONS, 2021](#)). The resulting junction-to-LSOA lookup is merged with the IMD 2019 LSOA-level dataset from MHCLG, which assigns each of England's **32,844 LSOAs** an IMD decile from 1 (most deprived) to 10 (least deprived) ([Somersetintelligence](#)) ([MHCLG, 2019](#)).

Ethnic composition join. ONS 2021 Census Table TS021 provides resident population by ethnic group at LSOA level ([ONS, 2022](#)). Joining this to SUMO junctions raises three issues. First, **legal defensibility under PSED**: section 149 of the Equality Act 2010 requires public authorities to have due regard to eliminating discrimination and advancing equality of opportunity across protected characteristics including race ([No5](#)) ([Equality Act 2010 s.149](#)). The Centre for Data Ethics and Innovation has stated explicitly that "public sector bodies have an even greater responsibility to consider and address bias when adopting algorithmic decision-making" ([Blog](#)) and that failure to test algorithms for discrimination may itself breach the PSED ([Blog](#)) ([CDEI, 2020](#)). Following *R (Bridges) v South Wales Police* [2020] EWCA Civ 1058, there is strong precedent that deploying algorithmic systems without assessing discriminatory impact is unlawful under PSED. ([Blog](#)) **Excluding ethnicity from the audit could therefore itself constitute PSED non-compliance.**

Second, **GDPR compliance**: the join operates on aggregate Census counts at LSOA level (minimum population ~1,500), not on individual-level data. No personal data is processed. The protocol imposes a **k-anonymity floor**: any LSOA with total population below 500 or fewer than 50 residents in any audited ethnic category is excluded from ethnic-group analysis, consistent with ONS disclosure control practices ([ONS, 2022](#)).

Third, **precedent**: LSOA-level demographic joins are established practice in UK transport equity research. Steinbach et al. linked STATS19 casualty data to LSOA-level census ethnicity and IMD deprivation for the London Road Safety Unit ([LSHTM Research Online](#)) ([ScienceDirect](#)) ([Steinbach et al., 2010, Health & Place](#)). Edwards et al. joined Hospital Episode Statistics to LSOA-level IMD for all of England, finding a rate ratio of 4.1 for serious pedestrian injuries in most versus least deprived areas ([PubMed](#)) ([Edwards et al., 2008, Archives of Disease in Childhood](#)). TfL's own 20 mph zone evaluation assigned IMD quintiles and census ethnic groups at LSOA level to casualty data ([TfL](#)) ([Grundy et al., 2008](#)). In the US, Kerr et al. joined satellite-derived NO₂ data to census-tract

demographics, finding persistent racial disparities in traffic-related pollution ([Kerr et al., 2021, PNAS](#)), and Chicago DOT's crash equity analysis linked fatality data to community-area demographics, finding that over half of traffic deaths involved Black residents despite comprising less than one-third of the population ([City of Chicago](#)) ([Chicago DOT, 2022](#)). The analysis explicitly reports **group-level patterns only** and does not infer individual ethnicity from traffic data.

§2.3 Statistical methodology

Baseline comparison. The primary test compares mean per-vehicle delay in the most-deprived LSOAs (IMD deciles 1-3) against the least-deprived (IMD deciles 8-10) using Welch's t -test and, given the right-skewed nature of delay distributions, the Mann-Whitney U test as a non-parametric alternative ([Technology Networks](#)) ([Mann & Whitney, 1947](#)). Effect size is reported as Cohen's d for the t -test and $r = Z/\sqrt{N}$ for Mann-Whitney, benchmarked against Cohen's conventions: small $d = 0.2$, medium $d = 0.5$, large $d = 0.8$ ([UCLA Statistics](#)) ([WordPress](#)) ([Cohen, 1988](#)).

Disparate impact ratio with bootstrap confidence intervals. The DIR is computed as the ratio of mean delay in least-deprived to most-deprived deciles. A **BCa bootstrap** with $B = 10,000$ resamples constructs a 95 % confidence interval; ([Wikipedia](#)) if the entire interval lies below 0.8, disparate impact is statistically confirmed ([Mills & Zandvakili, 1997, Journal of Applied Econometrics](#); [Morris & Lobsenz, 2000, Personnel Psychology](#)).

Regression with controls. Because SUMO is the data-generating process, the researcher controls traffic volume, demand profile, road geometry, and signal cycle length. This is a **decisive advantage for causal inference**: there is no unmeasured confounding in principle. Nevertheless, a regression of LSOA-level delay on IMD decile, controlling for SUMO-assigned traffic volume, road classification, and junction density, isolates the marginal contribution of deprivation to experienced delay and detects whether the AI controller amplifies or attenuates baseline inequalities.

Interventional experiments. The simulation permits **A/B counterfactual runs** with identical random seeds: one run with the fairness-constrained controller, one with the unconstrained controller. This makes counterfactual fairness in the Kusner et al. sense directly computable—the researcher observes the actual delay under both conditions for every vehicle, not an estimate. Causal estimands (average treatment effect of the fairness constraint, heterogeneous treatment effects by IMD decile) are recoverable without instrumental variables or matching.

Sample size and statistical power. The FHWA Traffic Analysis Toolbox Volume III recommends $N_{\min} = (t_{\alpha/2, N-1} \times s/e)^2$, where s is the standard deviation of the measure of effectiveness across initial runs and e is the tolerable error margin, typically converging at **10-20 replications** per scenario for 95 % CI with ± 5 % error ([Office of Operations](#)) ([FHWA-HOP-18-036](#)). For the between-LSOA comparison with 50-100 LSOAs and 30 replications per scenario, power at $\alpha =$

0.05 for a medium effect ($d = 0.5$) exceeds 0.80 when using all LSOAs in a Kruskal-Wallis framework rather than restricting to extreme deciles.

Multiple comparisons correction. The audit evaluates four metrics: one headline Gini, three triangulation. With $m = 4$ tests, **Holm-Bonferroni** is recommended: it controls the family-wise error rate (Wikipedia) while being uniformly more powerful than standard Bonferroni, and with only four comparisons the adjusted thresholds ($\alpha/4 = 0.0125$ for the most significant, $\alpha/3$, $\alpha/2$, α for subsequent) remain practical (Holm, 1979, *Scandinavian Journal of Statistics*; Noble, 2009, *Nature Biotechnology*).

§2.4 Real-world algorithmic equity audits — five case studies

UK Ofqual A-level algorithm (August 2020). Ofqual's Direct Centre Performance model moderated teacher-predicted grades using historical school-level distributions. (Wikipedia) **40 % of grades were downgraded,** (University of Bristol) with systematic bias: state school students and those in deprived areas received larger downward adjustments, while private schools with smaller cohorts retained teacher assessments. (British Educational Resear...) After mass protest, grades reverted to unmoderated teacher predictions. The head of Ofqual resigned. (British Educational Resear...) The case demonstrates how calibrating algorithms on historical institutional data encodes structural inequalities and underscores the need for pre-deployment equality impact assessment under PSED (Ofqual, 2020; Bristol CMM).

Netherlands SyRI welfare fraud detection (February 2020). The System Risk Indication linked data from tax, immigration, and municipal agencies to flag suspected benefit fraud. (LOC) The Hague District Court struck it down under Article 8 ECHR, (LOC) finding it lacked transparency (the risk model was secret), was disproportionate, (IAPP) and **deployed exclusively in low-income, high-immigrant neighbourhoods.** (AI Law Consultancy) The ruling established that public-sector algorithms must meet transparency, proportionality, and non-discrimination standards (Oxford Academic) (ECLI:NL:RBDHA:2020:1878; OHCHR, 2020).

Amazon AI recruiting tool (2014–2017). Amazon's resume-screening model, trained on a decade of predominantly male engineering hires, systematically penalised CVs containing the word "women's" and names of women's colleges. (Euronews) The tool was scrapped internally; Reuters reporting catalysed regulatory scrutiny, (American Civil Liberties Un...) contributing to NYC Local Law 144 requiring bias audits of automated employment tools (Dastin, 2018, Reuters).

Chicago Strategic Subject List (2012–2019). The SSL assigned risk scores to ~400,000 individuals predicting involvement in shootings. (Medium) A RAND evaluation found the system **did not reduce gun violence,** (Medium) (Wiley Online Library) while **85 % of highest-scored individuals were African-American men.** (Acgusa) (University of Chicago) Proxy variables (narcotics arrests, gang affiliation) encoded racialised policing patterns. (Chicago Magazine) (National Academies) The system was quietly decommissioned after sustained critique (RAND, 2016; DaViera et al., 2024, *AJCP*).

NYC Automated Decision Systems Task Force (2019). Established by Local Law 49 of 2018, the 18-member task force (City of New York) could not agree on a definition of automated decision systems, city agencies refused to disclose which algorithms they used, and the final report offered only broad recommendations with no enforceable audit framework. (S.T.O.P.) AI Now Institute's shadow report called it a failure. The case demonstrates that **voluntary review without statutory teeth cannot deliver algorithmic accountability** (NYC ADS Report, 2019; AI Now Shadow Report, 2019).

A notable absence: **no formal equity audit of traffic signal timing algorithms** was found in FHWA, US DOT, or TfL documentation. The Chicago Office of Inspector General's 2020 audit found that CDOT managed signals reactively via 311 complaints and aldermanic requests, (lgchicago) (lgchicago) systematically disadvantaging communities with less political capital (Chicago OIG, 2020). TfL conducts equality impact assessments for bus route changes (Transport for London) but not for signal timing algorithms. This gap is precisely what the present protocol addresses.

§2.5 Governance for the hypothetical real-world deployment

Three governance models present themselves. First, a **municipal oversight board** modelled on TfL's existing equality impact assessment process, in which the deploying authority self-audits quarterly and publishes results. This is the lowest-friction option but raises independence concerns, as demonstrated by the NYC ADS Task Force's failure. Second, a **third-party independent auditor**—a consultancy or academic institution contracted to execute the protocol against the blockchain log with read-only access. This mirrors the model used by the Dutch Algorithm Register, now hosting over 1,300 algorithms from 500+ government organisations (Digital Government) (Dutch Algorithm Register), and the Amsterdam register, which explicitly includes smart traffic lights in scope (OECD.AI, Amsterdam). Third, **civil society access**, where the audit log is published openly and advocacy groups may independently verify results, following Barcelona's seven-principle framework including (CIDOB) mandatory auditability and an Advisory Council on AI, Ethics and Digital Rights (Data-en-maatschappij) (Ajuntament de Barcelona) (Barcelona, 2021).

This protocol recommends a hybrid: the deploying authority executes the quarterly audit (§3) and registers the algorithm under the **UK Algorithmic Transparency Recording Standard** (ATRS), which became mandatory for central government in 2024 (UK Government) and follows a two-tier structure—Tier 1 plain-language summary for the public, Tier 2 detailed technical documentation for specialist audiences (GOV.UK) (GOV.UK ATRS Hub). An independent academic or civil-society auditor is granted read-only blockchain access to verify results annually. Quarterly audit reports are published as open data.

§2.6 Limits of equity audits

Five structural limitations constrain what any equity audit—simulated or real—can detect. **Induced non-travel:** the audit measures delay among those who travel; it cannot observe people

who chose not to travel because previous signal-induced delays made a journey unviable. **Mode shift:** if the signal policy pushes pedestrians into cars or onto different routes, the audit captures the new mode's delay but not the welfare loss of the forced switch. **Simpson's paradox:** LSOA-level aggregation can mask within-LSOA heterogeneity; a neighbourhood that appears equitably served in aggregate may contain a junction cluster that disproportionately delays one sub-group. **Temporal drift:** the population changes (ONS Census data is a 2021 snapshot), the model retrains (captured by the model-version field), and traffic patterns evolve. The quarterly cadence partially addresses model drift but cannot track inter-censal demographic change. **Sim-to-real validity:** SUMO's car-following models, demand generation, and pedestrian behaviour are approximations. Conclusions about relative fairness across architectural variants transfer well; absolute delay magnitudes do not. §4 elaborates this gap.

§3 Proposed audit protocol

This section specifies a numbered, implementable procedure for a quarterly equity audit of the AI traffic signal controller, executable against SUMO simulation output on the Elephant & Castle-Brixton corridor. All file paths, metric formulas, statistical tests, and pass/fail thresholds are defined.

Phase A — Data preparation

Step 1. Extract the SUMO network and define the study area. Import the OpenStreetMap extract for the bounding box [51.450°N, -0.125°W] to [51.495°N, -0.080°W] using `netconvert --osm-files lambeth_southwark.osm.xml -o corridor.net.xml --tls.guess-signals --tls.join` (SUMO OSM Import). Verify that the resulting network contains 25-100 signalised junctions (type `traffic_light` in `corridor.net.xml`). Export junction geo-coordinates: `netconvert --sumo-net-file corridor.net.xml --plain-output-prefix plain --proj.plain-geo` (SUMO Geo-Coordinates). Output: `junctions.csv` with columns `[junction_id, lon, lat]`.

Step 2. Perform the spatial join to LSOAs. Download 2021 LSOA boundaries (BGC, GeoJSON) from the ONS Open Geography Portal (ONS, 2021). Using GeoPandas, point-in-polygon join each junction to its containing LSOA. Output: `junction_lsoa_lookup.csv` with columns `[junction_id, lsoa_code, lsoa_name]`.

Step 3. Join IMD 2019 deciles. Download the IMD 2019 LSOA-level dataset (MHCLG, 2019). Merge on `lsoa_code`. Output: `junction_lsoa_imd.csv` adding column `[imd_decile]` (1 = most deprived, 10 = least deprived).

Step 4. Join ONS 2021 Census ethnic group data. Download Census Table TS021 at LSOA level (ONS, 2022). Merge on `lsoa_code`. Apply the **k-anonymity floor**: exclude any LSOA with total usual residents < 500 or with fewer than 50 residents in any of the five high-level ethnic

categories (White; Mixed/Multiple; Asian, Asian British or Asian Welsh; Black, Black British, Black Welsh, Caribbean or African; Other). Record the number of LSOAs excluded and the reason. Compute for each remaining LSOA: proportions in each of the five categories. Output: `lsoa_demographics.csv`. **Explicit constraint:** this analysis reports group-level patterns only and does not infer individual ethnicity from traffic data. Inclusion of ethnicity is justified as a PSED obligation: s.149 of the Equality Act 2010 requires public authorities to have due regard to advancing equality of opportunity across protected characteristics including race, and the CDEI has stated that failing to test algorithms for discriminatory impact may itself breach the duty (CDEI, 2020).

Step 5. Assign vehicles to LSOAs. For each completed vehicle trip in the SUMO TripInfo output (`--tripinfo-output trips.xml`), identify the junction nearest to the trip's origin edge. Assign the vehicle to the LSOA of that junction via `junction_lsoa_lookup.csv`. Extract per-vehicle fields: `timeLoss` (seconds of delay relative to free-flow) and `waitingTime` (seconds at speed < 0.1 m/s) (SUMO Output). For pedestrian trips, extract `duration` and compute pedestrian delay as `duration - (trip_distance / 1.2)`, where 1.2 m/s is the NCHRP 969 lower-bound walking speed for mobility-impaired adults. Output: `vehicle_delays.csv` and `pedestrian_delays.csv`.

Phase B — Simulation execution

****Step 6. Define the experimental design.**** Each quarterly audit executes two conditions—**fairness-constrained controller** (FC) and **unconstrained controller** (UC)—each replicated $R = 30$ times with seeds 1, 2, . . . , 30, using SUMO's `--seed` flag ([SUMO Randomness] (<https://sumo.dlr.de/docs/Simulation/Randomness.html>)). The demand profile is held constant across conditions: a 3-hour AM peak (07:00–10:00) generated from the same OD matrix. Total runs per quarter: $2 \times 30 = 60$. FHWA guidance indicates 10–20 replications suffice for $\pm 5\%$ error at 95 % confidence; 30 provides a margin for power at small effect sizes ([FHWA-HOP-18-036] (<https://ops.fhwa.dot.gov/publications/fhwahop18036/index.htm>)).

Step 7. Execute runs and collect logs. For each run $r \in \{1, \dots, 30\}$ and condition $c \in \{\text{FC}, \text{UC}\}$: launch SUMO with `--seed r --tripinfo-output trips_c_r.xml --summary-output summary_c_r.xml`. Simultaneously, the AI controller writes per-decision records to the blockchain log. After each run, parse `trips_c_r.xml` and append to `vehicle_delays.csv` with columns `[condition, seed, vehicle_id, origin_lsoa, timeLoss, waitingTime]`.

****Step 8. Aggregate to LSOA level.**** For each (condition, seed, LSOA) triple, compute: mean per-vehicle `timeLoss` ($\bar{d}_{c,r,i}$), median `timeLoss`, 95th-percentile `timeLoss`, and mean pedestrian delay. Then average across seeds: $\bar{d}_{c,i} = \frac{1}{R} \sum_r \bar{d}_{c,r,i}$. Output: `lsoa_delay_summary.csv`.

Phase C — Metric computation

Step 9. Compute the headline Gini coefficient. For the FC condition, compute:

$$G_{FC} = \frac{1}{2n^2\bar{d}_{FC}} \sum_{i=1}^n \sum_{j=1}^n |\bar{d}_{FC,i} - \bar{d}_{FC,j}|$$

where n is the number of LSOAs in the study area and $\bar{d}_{FC}$ is the grand mean. Apply the bias correction $G^* = G \times n/(n - 1)$. Repeat for UC. Construct **BCa bootstrap 95 % confidence intervals** with $B = 10,000$ resamples (Dixon et al., 1987, *Ecology*; R: `(DescTools::Gini(x, conf.level=0.95)) (DescTools)`). Output: `[gini_results.csv]` with columns `[condition, gini, ci_lower, ci_upper]`. Generate Lorenz curves for both conditions on a single plot.

Pass/fail threshold for Gini: Drawing on transport accessibility literature where Gini > 0.3 is flagged as concerning and > 0.4 as significant inequality (Rao et al., 2022), the protocol sets $G < 0.30$ as the pass threshold for the FC condition. If $G_{FC} \geq 0.30$, the audit flags a fairness concern. If $G_{FC} \geq 0.40$, the audit flags significant inequality requiring immediate review.

Step 10. Compute the three triangulation metrics.

10a. Rawlsian max-min. Compute $M_c = \text{mean}(\bar{d}_{c,i} : \text{LSOA}_i \in \text{IMD deciles 1-3})$ for each condition. Report the absolute value and the ratio M_{FC}/M_{UC} . A ratio below 1.0 indicates the fairness constraint improves the worst-off group. No fixed pass/fail; report alongside the system-wide mean to contextualise.

10b. Disparate impact ratio. Compute $\text{DIR} = \bar{d}_{\text{IMD 8-10}}/\bar{d}_{\text{IMD 1-3}}$ for the FC condition, where a lower ratio indicates the most-deprived group waits longer. Construct a BCa bootstrap 95 % CI with $B = 10,000$. **Pass/fail: $\text{DIR} \geq 0.80$ ** (the four-fifths rule adapted from [29 CFR §1607.4D](https://www.ecfr.gov/current/title-29/subtitle-B/chapter-XIV/part-1607)). If the entire CI falls below 0.80, the audit concludes statistically significant disparate impact.

10c. Pedestrian delay parity. Compute $P_c = \text{mean pedestrian delay in IMD deciles 1-3} / \text{mean pedestrian delay in IMD deciles 8-10}$ for the FC condition. Following the NCHRP 969 principle that pedestrian delay should not be treated as a lesser concern ([NCHRP, 2022](https://doi.org/10.17226/26491)) and Zhang et al.'s human-centric framework ([Zhang et al., 2026](https://arxiv.org/abs/2602.02959)), the **pass threshold is $P \leq 1.25$ **: pedestrians in the most-deprived areas should not wait more than 25 % longer than those in the least-deprived areas.

Step 11. Ethnic-group disaggregation. For each of the five ONS high-level ethnic categories, identify LSOAs where that group constitutes ≥ 30 % of the population (a conventional threshold for "majority-minority" analysis). Compute mean delay for the set of LSOAs associated with each ethnic group. Report pairwise DIR values between each group and the group with the lowest delay. Flag any pairwise $\text{DIR} < 0.80$. This analysis is supplementary; the headline and triangulation metrics remain IMD-based.

Phase D — Statistical testing

Step 12. Difference-in-means tests. For the FC condition, test H_0 : mean delay in IMD deciles 1–3 equals mean delay in IMD deciles 8–10. Compute Welch's t -test and Mann-Whitney U ([Mann & Whitney, 1947](#)). Report p -values, Cohen's d , and $r = Z/\sqrt{N}$.

Step 13. Regression analysis. Regress $\bar{d}_{FC,i}$ on IMD decile (as a continuous variable), controlling for SUMO-assigned junction traffic volume, road classification, and number of signal phases. Use heteroskedasticity-robust standard errors clustered by LSOA. Report the coefficient on IMD decile with 95 % CI. A significant positive coefficient indicates that more-deprived LSOAs experience higher delay after controlling for traffic characteristics.

****Step 14. Interventional comparison (FC vs UC).**** For each LSOA i , compute $\Delta_i = \bar{d}_{UC,i} - \bar{d}_{FC,i}$. Test whether Δ_i is systematically larger for more-deprived LSOAs using the same regression framework (Step 13) with Δ_i as the dependent variable. A significant positive coefficient on IMD decile indicates the fairness constraint disproportionately benefits deprived neighbourhoods—the desired outcome.

Step 15. Apply Holm-Bonferroni correction. Collect the four p -values from the headline Gini test (testing $H_0 : G = 0$, i.e., no inequality) and the three triangulation metrics' pass/fail tests. Sort ascending. Compare each against $\alpha/(4 - k + 1)$ for rank k ([Holm, 1979](#)). Report corrected significance for all four metrics in a single summary table.

Phase E — Reporting and blockchain logging

Step 16. Produce the quarterly audit report. The report is a structured document containing:

- **Summary dashboard:** four-metric table (Gini, Rawlsian max-min, DIR, pedestrian parity) with values, CIs, pass/fail status, and trend arrows comparing to the previous quarter.
- **Lorenz curve plot:** FC vs UC conditions.
- **Choropleth map:** mean LSOA-level delay overlaid on the Elephant & Castle–Brixton corridor, shaded by IMD decile.
- **Ethnic-group supplementary table:** mean delay by LSOA ethnic composition cluster, with pairwise DIRs.
- **Regression coefficient table:** IMD decile effect with controls.
- **Interventional effect table:** FC vs UC differential by IMD decile.
- **Data provenance statement:** model version, SUMO version, demand scenario hash, seed range, number of LSOAs included/excluded (with k -anonymity floor reasons), Census vintage, IMD vintage.

Step 17. Hash and log the report to the blockchain. Compute SHA-256 of the audit report PDF. Write a blockchain transaction containing: `[quarter_id, report_hash, gini_value, dir_value, pedestrian_parity_value, rawlsian_value, model_version, auditor_id, timestamp]`. The append-only ledger ensures that no historical audit result can be retroactively altered.

Step 18. Publish. Deposit the full report and underlying `lsoa_delay_summary.csv` in the deploying authority's open-data portal and register the algorithm version under the UK ATRS Tier 1 and Tier 2 templates (GOV.UK ATRS). Notify the independent auditor (if appointed) that blockchain-verified data is available for annual verification.

Quarterly cadence justification

The quarterly cadence aligns with three cycles. First, **model retraining**: AI signal controllers are typically retrained or fine-tuned on a monthly-to-quarterly basis; auditing after each retraining epoch captures performance changes. Second, **seasonal demand variation**: quarterly audits span all four seasons within a year, enabling detection of fairness patterns that manifest only under winter darkness, school-term traffic, or summer leisure patterns. Third, **governance precedent**: the Amsterdam Algorithm Register expects periodic review of registered algorithms, and TfL reports performance metrics quarterly. The quarterly cadence is frequent enough to detect drift before harm accumulates, yet infrequent enough to be sustainable on a developer laptop running SUMO (60 runs $\times$ ~15 minutes each = ~15 hours of computation per quarter).

Worked example: what the researcher produces

On the Elephant & Castle–Brixton corridor, the researcher produces the following artefacts each quarter:

1. `corridor.net.xml` — SUMO network (stable across quarters unless the road graph changes).
2. `junction_lsoa_imd.csv` — ~60–80 junctions mapped to ~40–60 LSOAs with IMD deciles.
3. `lsoa_demographics.csv` — ethnic composition for each LSOA passing the k-anonymity floor.
4. 60 TripInfo XML files (30 FC seeds + 30 UC seeds).
5. `lsoa_delay_summary.csv` — aggregated delay statistics.
6. `gini_results.csv` — headline Gini with bootstrap CIs.
7. `audit_report_Q[n]_[year].pdf` — the structured report.
8. Blockchain transaction receipt — immutable hash-anchored record.

The expected Gini for an unconstrained controller on this corridor, given the sharp IMD gradient between Elephant & Castle (deciles 1–2) and Herne Hill/Dulwich (deciles 7–9), is likely in the

range 0.15–0.35 based on analogous transport accessibility Gini values. The fairness-constrained controller should reduce this. A successful audit shows $G_{FC} < 0.30$, $DIR \geq 0.80$, and pedestrian parity ≤ 1.25 .

§4 Explicit limitations of the simulation-based demonstration

What the SUMO audit can support

The protocol supports four categories of claim. First, **relative fairness comparison**: the interventional design (FC vs UC, identical seeds) provides unambiguous evidence of whether the fairness constraint improves equity, and by how much, across every metric. This is a causal claim, recoverable because the simulator is the data-generating process with no unmeasured confounding. Second, **existence of a procedure**: the audit demonstrates that a concrete, implementable, statistically rigorous equity audit protocol for AI traffic signal control exists and can be executed, filling the gap identified in §2.4 where no FHWA, US DOT, or TfL signal-timing equity audit was found. Third, **statistical power calibration**: the 30-replication design with known variance enables precise power calculations, informing the minimum sample size for a future real-world audit. Fourth, **sensitivity analysis**: the researcher can systematically vary demand levels, network topology, and fairness-constraint parameters to map the conditions under which equity degrades.

What would require real-world deployment data

Four conclusions lie beyond the reach of simulation. **Induced non-travel** — detecting people who did not travel because the signal system made journeys unviable — requires longitudinal travel-diary or mobile-phone mobility data from before and after deployment. **Mode-shift harms** — the welfare loss when a pedestrian switches to a car because of hostile signal timing — require revealed-preference data on mode choice. **Actual demographic incidence** — confirming that the LSOA-level Census proportions accurately represent who actually uses each junction — requires intercept surveys or anonymised sensor data. **Regulatory standing** — whether a SUMO-based audit satisfies a court or regulator in the event of a legal challenge under the Equality Act 2010 — is an untested legal question; the protocol's value lies in demonstrating due diligence and methodological rigour, not in pre-empting judicial interpretation.

The sim-to-real gap for traffic signal AI

SUMO's car-following models (Krauss, IDM) and lane-changing models are calibrated to aggregate traffic behaviour but do not capture every real-world interaction: pedestrian non-compliance with signals, informal road-crossing, cyclist behaviour at advanced stop lines, and the effects of weather and road surface on travel speed are all approximated or absent. Demand matrices derived from strategic models carry their own biases. Junction geometries imported from OpenStreetMap may contain errors in lane counts or turn movements. **Relative**

comparisons within SUMO (FC vs UC, LSOA A vs LSOA B) are robust to these limitations because the biases are shared across conditions. Absolute delay values are not directly transferable. Any deployment following a successful simulation audit should therefore include a **calibration phase** in which SUMO delays are validated against real-world detector data from the first weeks of operation, and the audit thresholds (Gini < 0.30 , DIR ≥ 0.80 , pedestrian parity ≤ 1.25) are reviewed in light of observed baseline variance.

Conclusion

This chapter has specified a quarterly equity audit protocol that transforms the blockchain-logged output of an AI traffic signal controller into actionable fairness evidence. The protocol's distinctive contribution is threefold: it exploits SUMO's interventional capability to recover causal fairness estimands that observational audits cannot; it operationalises the UK Public Sector Equality Duty by joining simulated traffic outcomes to real IMD and Census data at LSOA level with explicit privacy safeguards; and it anchors every quarterly result to an immutable blockchain record, creating a tamper-evident audit trail. The Gini headline metric, triangulated by Rawlsian max-min delay, disparate impact ratio, and pedestrian delay parity, provides a multi-dimensional fairness assessment that no existing traffic signal audit—simulated or real—has previously specified to this level of procedural detail. The protocol is immediately executable on a developer laptop against the Elephant & Castle–Brixton corridor and is designed to scale, with minimal modification, to the ~6,400 signalised junctions of the London network should real-world deployment proceed.